Evaluation Methodology

4.1 Dataset Construction and Experimental Design

To rigorously evaluate the proposed hallucination detection system, a structured evaluation dataset consisting of 100 questions was carefully constructed to provide comprehensive coverage of the information contained within the system's designated knowledge context. The questions were designed to probe a wide range of factual, relational, and inferential claims that a large language model (LLM) might be expected to address, thereby ensuring that the evaluation captures diverse response patterns and potential failure modes. Each question was independently submitted to four distinct large language models, and their generated responses were subsequently passed through the hallucination detection pipeline for automated analysis.

4.2 Ground Truth Establishment

To establish a reliable ground truth for evaluation, all model-generated responses underwent systematic manual inspection. Each response was carefully reviewed by the investigator, and any content deemed to constitute a hallucination — defined as a statement that is factually inconsistent with, unsupported by, or contradictory to the information present in the knowledge context — was identified and recorded. This manual annotation process served as the reference standard against which the automated system's outputs were subsequently assessed.

4.3 Triple Extraction and Hallucination Classification

The detection pipeline operates by decomposing each model response into semantic triples, which represent atomic subject–predicate–object units of information. Each extracted triple was then systematically evaluated against the knowledge context and classified into one of the following categories:

- **Rule Violation: Triples that contradict explicit constraints or logical rules defined within the knowledge context.**

- **Predicate Mismatch: Triples in which the relationship between entities is incorrectly stated or misrepresented.**

- **Fabricated Fact: Triples that assert information with no basis or correspondence within the knowledge context.**

- **Logical Inconsistency: Triples that are internally contradictory or inconsistent with other statements within the same response.**

- **Unverifiable: Triples that could neither be confirmed nor refuted based on the available knowledge context. These were retained in the evaluation to avoid inflating performance metrics.**

4.4 Evaluation Metrics

Two primary metrics were employed to quantify system performance: the Hallucination Rate and the Detection F1-Score.

Hallucination Rate. **The hallucination rate for each model was computed as the proportion of hallucinated triples relative to the total number of extracted triples, including those classified as unverifiable. This metric provides a model-level indicator of the frequency of erroneous or unsupported content generation, and is formally defined as:**

Hallucination Rate = (Total Hallucinated Triples) / (Total Extracted Triples incl. Unverifiable)

Detection F1-Score. **To assess the accuracy of the automated detection system, its output was compared against the manually established ground truth using the F1-Score, a standard metric that balances precision and recall. The F1-Score is particularly appropriate in this context as it accounts for both false positives — triples incorrectly flagged as hallucinations — and false negatives — hallucinations present in the ground truth that the system failed to detect. The metric is formally defined as:**

$$F1 = (2 \times TP) / (2 \times TP + FP + FN) \equiv (2 \times TP) / (TP + FP + H_m)$$

where TP denotes the number of hallucinated triples correctly identified by the system, FP denotes the number of triples incorrectly flagged as hallucinations, FN denotes the number of hallucinations present in the ground truth that the system failed to detect, and H_m denotes the total number of hallucinations identified through manual review (i.e., $H_m = TP + FN$). It is noted that the formulation using H_m is algebraically equivalent to the standard F1-Score, and is adopted here to more intuitively reflect the relationship between the system's detections and the manually verified ground truth.

4.5 Evaluation Framework Summary

The evaluation framework described above enables a dual-level analysis: at the model level, the hallucination rate provides insight into the propensity of each LLM to generate unsupported or erroneous content; at the system level, the F1-Score quantifies the reliability and effectiveness of the automated detection pipeline relative to human judgment. Together, these metrics offer a comprehensive and reproducible basis for evaluating the proposed system's performance across diverse models and response types.